

Statistics Summative Project

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Abstract

Placeholder text. This is where the abstract is going to be. I'm going to write it last, as per standard procedure.

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1 Hypotheses

We seek to prove that some parts of this data, and some pairs, have association. By performing this test, we attempt to find associations between abstract data values and tests.

H_0 = There is no association within the data.

H_a = There is an association, or multiple, within the data.

To reject the null hypothesis, we must prove that the alternative hypothesis is correct. In order to do so, we must use the χ^2 values and their associated p -values, as well as graphs and charts depicting these data, to independently analyze them if the χ^2 tests do not provide enough information to support the alternative hypothesis.

1.1 Specific general hypotheses

Some general hypotheses we have about this data are:

- There are associations between philosophical outlook and school.
- There are associations between randomness and age due to maturity.
- MIC will have more STEM-oriented hobbies due to its emphasis on STEM and its attractiveness to STEM enthusiasts.
- SWHS will play more sports on average due to dedicated programs.
- MIC will have a significantly higher proportion of students eating school lunch due to a higher quality of school lunch served.
- There will be a significant difference in weighted GPA between the two schools.
- MIC will have a higher rate of college attendance.
- There will be a significant difference in variability between the two academic environments.

To test these hypotheses, we must perform tests in order to determine their significance, such as two-sample significance tests for means, χ^2 significance tests, and we will analyze the p-values at $\alpha = 0.05$ to determine significance.